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AUTOMOTIVE



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PROJECT OVERVIEW





INTRODUCTION

Toyota Motor Corporation is a global automotive company earning revenue from vehicle sales and after-sales services through its worldwide dealerships.

Integrating sales, inventory, supply chain, and customer data into a centralized data warehouse enables effective performance monitoring and better strategic decision-making.





OBJECTIVES

- Design a dimensional Data Warehouse
- Support Business Intelligence (BI)
- Integrate sales, service & inventory data
- Enable KPI & dealership performance monitoring
- Analyze customer behavior
- Support strategic business growth





STAKEHOLDERS

- **Sales Managers** – Monitor revenue & sales performance
- **Dealership Managers** – Evaluate operations & profitability
- **Supply Chain Managers** – Manage inventory & stock flow
- **Service Managers** – Track service efficiency & warranty issues
- **Executives** – Oversee strategic & overall performance



BUSINESS QUESTIONS

- Which models and regions generate the highest revenue?
- Which dealerships achieve the best profitability?
- How efficient is inventory turnover across regions?
- What are the service cost and frequency per vehicle model?
- Which customers are most likely to repurchase?





KEY PERFORMANCE INDICATORS (KPIs)

1. Sales Performance

- Total Revenue
- Sales Volume
- Average Sales Price (ASP)
- Discount & Promotion Impact

2. Service Efficiency

- Service Revenue
- Labor Productivity
- Vehicle Downtime
- Warranty Ratio

3. Inventory Management

- Inventory Asset Value
- Stock Availability Rate
- Inventory Aging
- Sell-through Rate





DATA SOURCE

Primary Data Source

- Toyota Used Cars Dataset (Kaggle)
Vehicle specifications & pricing
- Car Sales Dataset (Kaggle)
Sales transactions, dealership & customer data



Additional Generated Data

- Generated dealership, customer, inventory & service data
- Created due to privacy constraints
- Used to simulate realistic automotive operations



MEASURES & DIMENSIONS OVERVIEW

Fact Tables

- Sales Fact – Stores vehicle sales transactions.
- Service Fact – Stores maintenance and repair transactions.
- Inventory Fact – Stores vehicle stock movement and inventory records.

Dimension Tables

- Date – Time-based analysis
- Car – Vehicle characteristics
- Dealership – Branch performance
- Customer – Segmentation & behavior
- Salesperson – Sales performance
- Promotion – Marketing impact
- Payment Method – Payment behavior
- Technician – Service efficiency
- Service Type – Maintenance category
- Supplier – Procurement analysis
- Location – Warehouse tracking

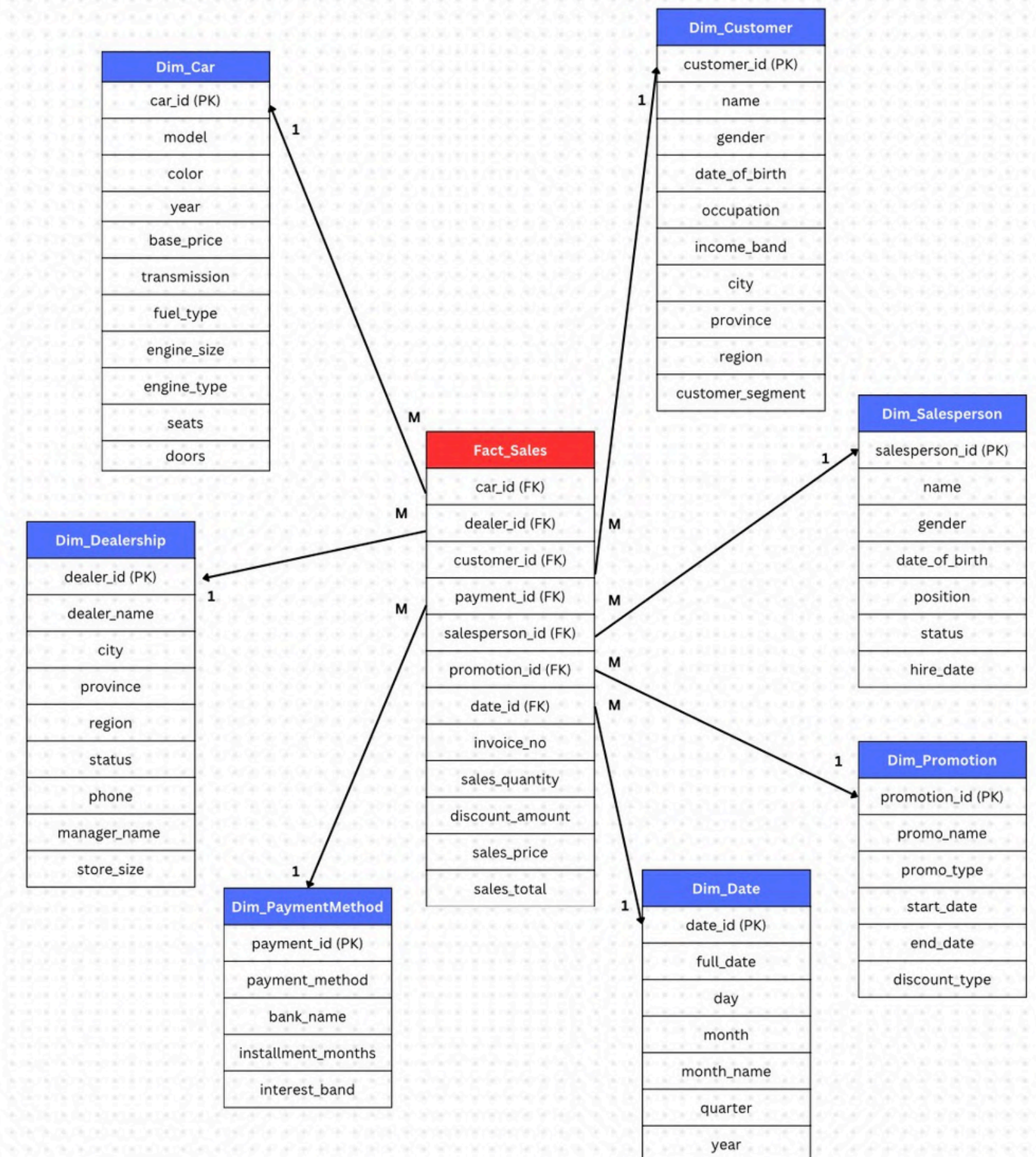


Fact Sales

- Captures granular sales transactions (units, price, discount)
- Integrates key dimensions (Date, Car, Customer)
- Enables regional & model performance analysis
- Supports strategic decisions & profitability tracking

DIMENSIONAL MODELING (GRAIN)

Sales Fact: One car sale transaction for one customer, on one date, at one dealership, handled by one salesperson, with one promotion and one payment method.

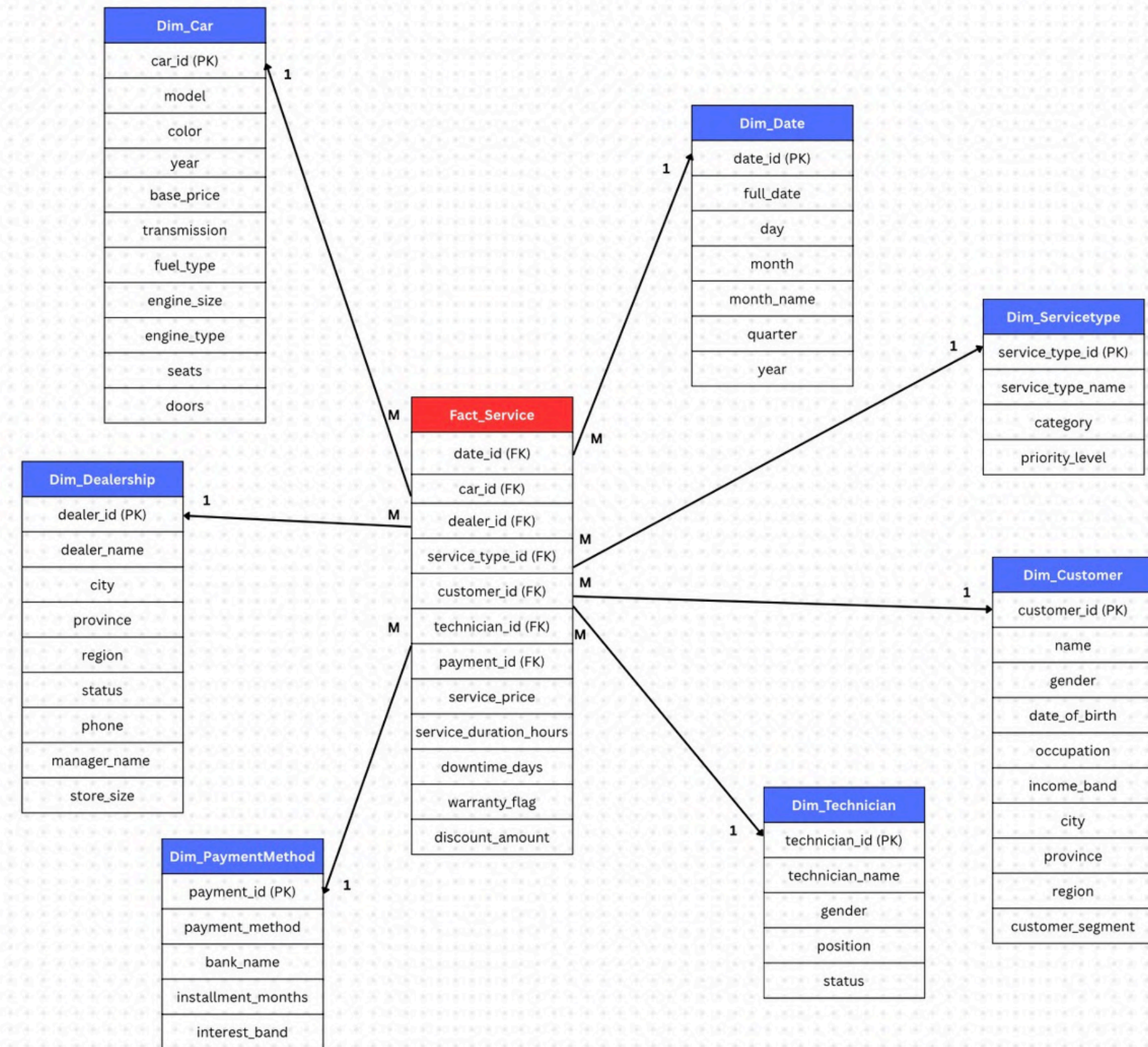


Fact Service

- Tracks service revenue, labor time & downtime
- Evaluates technician productivity
- Analyzes service type demand
- Supports resource optimization & service quality improvement

DIMENSIONAL MODELING (GRAIN)

Service Fact: One service transaction for one car, performed for one customer, on one date, at one dealership, handled by one technician, for one service type, with one payment method.

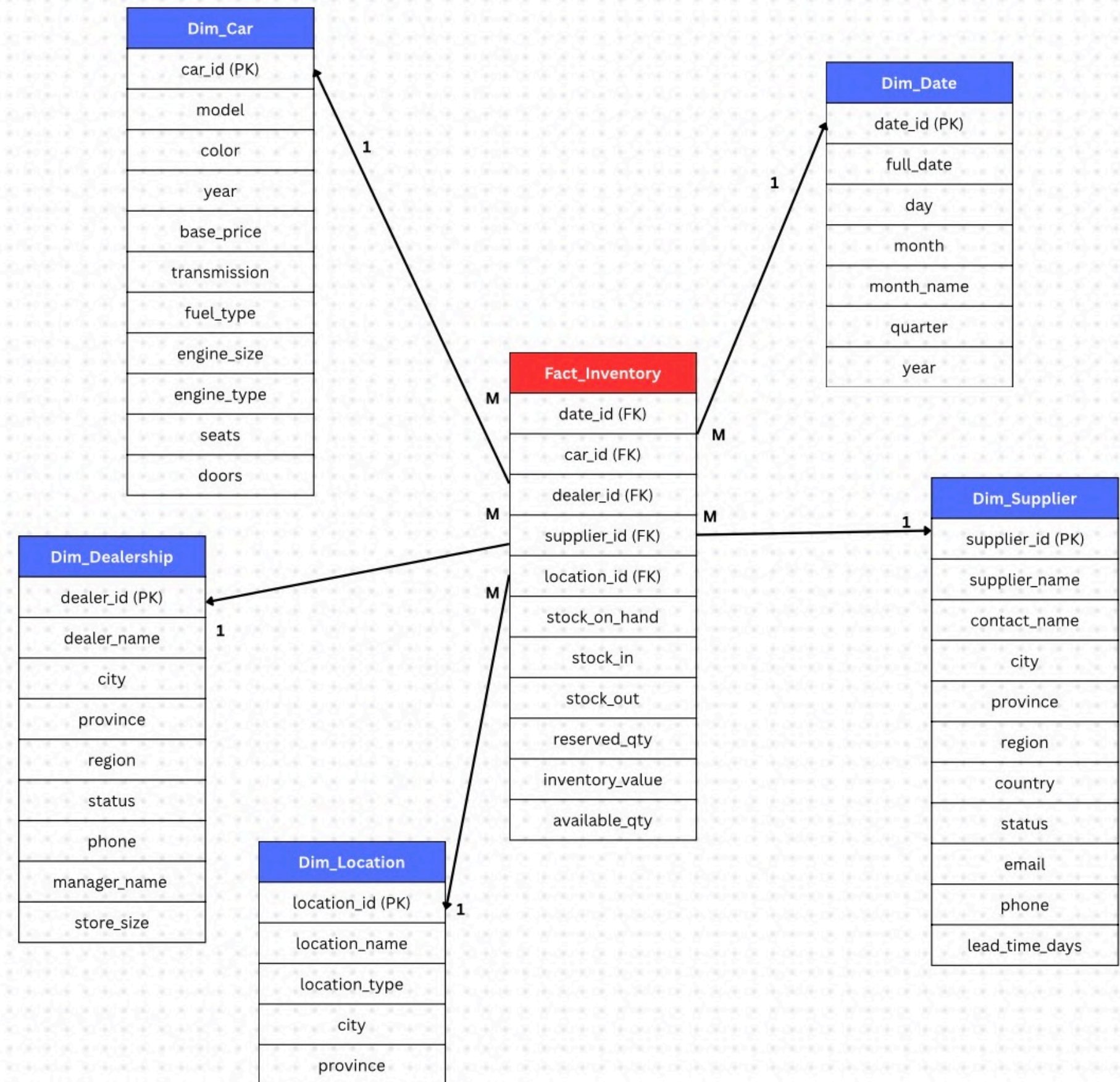


Fact Inventory

- Tracks stock levels & movements (in/out/available)
- Improves supply chain visibility
- Identifies slow-moving inventory
- Evaluates supplier reliability & reduces carrying costs

DIMENSIONAL MODELING (GRAIN)

Inventory Fact: One inventory record for one car, at one dealership, in one specific location, supplied by one supplier, on one date.



Data Dictionary

Table	Attribute	Type	Format	Key	Reference	Description
Fact_Sales	date_id	INT	11	FK	Dim_Date (date_id)	Links to Dim_Date
Fact_Sales	car_id	INT	11	FK	Dim_Car(car_id)	Links to Dim_Car
Fact_Sales	dealer_id	INT	11	FK	Dim_Dealership (dealer_id)	Links to Dim_Dealership
Fact_Sales	customer_id	INT	11	FK	Dim_Customer (customer_id)	Links to Dim_Customer
Fact_Sales	payment_id	INT	11	FK	Dim_PaymentMethod (payment_id)	Links to Dim_PaymentMethod
Fact_Sales	salesperson_id	INT	11	FK	Dim_Salesperson (salesperson_id)	Links to Dim_Salesperson
Fact_Sales	promotion_id	INT	11	FK	Dim_Promotion (promotion_id)	Links to Dim_Promotion
Fact_Sales	invoice_no	VARCHAR	30	-	-	Unique transaction number (degenerate dimension)
Fact_Sales	sales_quantity	INT	11	-	-	Units sold
Fact_Sales	discount_amount	DECIMAL	10,2	-	-	Total discount applied to the sale
Fact_Sales	sales_price	DECIMAL	10,2	-	-	Unit selling price after discount
Fact_Sales	sales_total	DECIMAL	12,2	-	-	Total revenue (sales_price × sales_quantity)

Data Dictionary

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Car	car_id	INT	11	PK	-	Surrogate key for car
Dim_Car	model	VARCHAR	50	-	-	Car model
Dim_Car	color	VARCHAR	50	-	-	Color of car
Dim_Car	year	INT	4	-	-	Manufacture year
Dim_Car	base_price	DECIMAL	10,2	-	-	Standard price before discount
Dim_Car	transmission	VARCHAR	20	-	-	Transmission type
Dim_Car	fuel_type	VARCHAR	20	-	-	Fuel type
Dim_Car	engine_size	DECIMAL	3,1	-	-	Engine capacity
Dim_Car	engine_type	VARCHAR	50	-	-	Engine configuration
Dim_Car	seats	INT		-	-	Number of seats
Dim_Car	doors	INT		-	-	Number of doors



BENEFITS

Integrated Business View

- Unified sales, service & inventory analysis
- Holistic dealership performance evaluation

KPI & Trend Monitoring

- Efficient revenue & stock tracking
- Time-based analysis (daily → yearly)

Scalable & Flexible Design

- Easy to add new dimensions & fact tables
- Supports long-term BI expansion



CHALLENGES

Data Quality Issues

- Missing & inconsistent public data
- High ETL cleaning effort

Synthetic Data Usage

- Simulated operational environment
- May not fully reflect real-world complexity

ETL Integration Complexity

- Schema mapping & key alignment
- Granularity consistency management

LIMITATIONS

- Limited real operational data
- No real-time processing (batch ETL only)
- Simplified customer dimension





CONCLUSION

- Unified sales, service & inventory data
- Eliminated data fragmentation
- Enabled KPI monitoring & performance evaluation
- Supported strategic decision-making
- Enhanced operational efficiency & long-term growth

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Q & A SESSION



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